Lost in Transmission: An Analysis of GNSS Robustness

Letizia Capitanio, Melissa Moioli,
Alberto Ovena, Tiziano Radicchi
Group: `void (*antani)(int, int)`
Politecnico di Torino
Italy

ABSTRACT

GNSS architectures are continuously challenged by external signal interference and malicious threats. This report aims to analyze positioning systems' resilience in various contexts, from ideal scenarios to environments with potential sources of interference, as well as the effects of simulated spoofing scenarios. The obtained results suggest that measurements can vary greatly among different GNSS chipsets even in ideal scenarios, while physical obstacles and spoofing attacks can effectively undermine GNSS resilience.

1 METHODS

The experiment's goal is to analyze GNSSs' performance in different scenarios, differing in (i) physical obstacles, (ii) time of the day, (iii) weather conditions and (iv) sources of interferences, both on the front-end and with simulated cyberspoofing.

The experimental setup consists of three physical Android devices, listed in Table 1. On all the devices, the GnssLogger app is used exclusively to track raw measurements [3], which are then analyzed via the GnssMeasurement MATLAB script [1].

Brand	Model	GNSS Chipset
Google	Pixel 7a	Broadcom BCM4776 series
Samsung	A51	Samsung Exynos 9611
Redmi	Note 7	Qualcomm SDM660

Table 1: Devices.

Scenario	Moving	Day Time	Weather
Line of Sight	No	15:00	Sunny
Indoor	No	16:00	Sunny
Subway	Yes	20:00	Cloudy
Car	Yes	11:00-21:00	Cloudy
DVB-T Antenna	No	22:00	Cloudy

Table 2: Measured scenarios.

Each scenario in Table 2 is sampled using the devices in Table 1. All scenarios are tested with and without *energy saving* modes on all devices, and results are later discussed[1]. Each measurement lasts 5 minutes. For each device-scenario pair, 4 samples are collected.

As regards the spoofing activity, a spoofing simulation was performed via the MATLAB script. Location spoofing is applied both with and without an additional delay in transmission. Furthermore, the script is modified to apply delays to a chosen subset of satellites, and the results are explored. All scripts are available in the repository.

The remainder of the report is structured as follows. Section 2 firstly focuses on performance variance among different devices, later presenting how positioning is affected in the case of simulated cyberspoofing, and finally offering empirical comparisons in more hostile environments. Section 3 concludes the performed experiments and presents possible additional evolutions for future work.

2 RESULTS AND DISCUSSION

2.1 Performance Analysis

A performance analysis is proposed, adopting an ideal scenario with sunny weather and no major natural/architectural obstructions. The different chipsets at our disposal are leveraged to demonstrate how performance can vary among different devices, even in ideal conditions.

In the *Line of Sight* scenario, all chipsets demonstrate similar high-level behaviour in the distribution of positions, averaging 10.78m variance for 50% of the distribution without power saving mode. Enabling said mode slightly increases the variance, bringing it to 11.28m. The most accurate measurements are produced by the Redmi Note 7 (with a variance of 7.6m with power save disabled and of 9.5m when enabled), whereas the Google Pixel 7a reveals high instability (ranging from a variance of 7.0m to one of 19.3m with power save disabled in the same location and within a time difference of minutes, even once improving with power save enabled), the worst example of which can be appreciated in Figure 1. The difference in the two measurements for the Google Pixel 7a is also supported by higher Horizontal Dilution of Precision (HDOP) for the worst sample, although the number of satellites is higher for the best one. This phenomenon is observable in Appendix Figure 4. Speaking of satellites, the Samsung A51 is constantly able to receive more (often reaching 10 and rarely dropping below 8) and offers the lowest average HDOP. The measurements differ by a 10-minute offset at most, maintaining the same position on the ground. This does not drastically affect the satellite geometry, allowing for these comparisons.

The analysis of this behaviour advances the hypothesis that a GNSS chipset which is integrated in the device's SoC offers overall better performance with respect to featuring a separate one. Indeed, both the Samsung A51 and the Redmi Note 7 (which feature integrated GNSS chipsets) deliver measurements which are both more stable and more accurate, in contrast with the far more unbalanced readings of the Google Pixel 7a. However, many other factors (e.g., firmware, antenna quality) can affect performance, as previous research suggested [6], [7], [2].

[1]The repository containing the used dataset, additional plots, and scripts is publicly available here.

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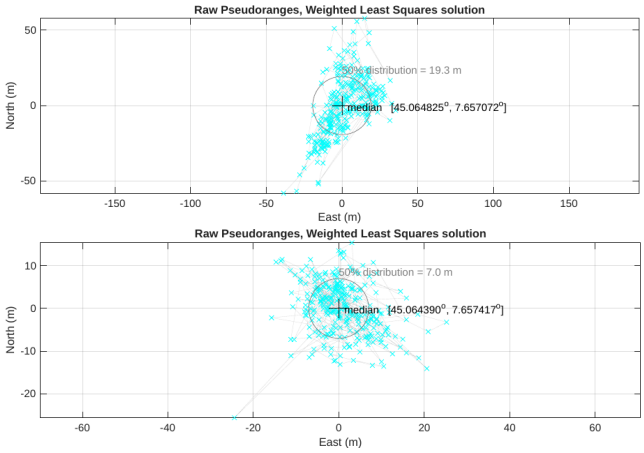

Figure 1: Variance of positions for the worst (top) and best (bottom) sample recorded by the Google Pixel 7a, power save disabled, in the *Line of Sight* scenario.

2.2 Cyberspoofing Effects

Starting from the baseline measurements of the *Line of Sight* scenario, this section proposes an evaluation of the impact of GNSS cyberspoofing techniques, specifically (i) direct position alteration and (ii) signal delay manipulation. By analyzing the statistical and geometric discrepancies between legitimate and spoofed measurements, we assess vulnerabilities and propose possible countermeasures to mitigate their effects.

2.2.1 Position Spoofing. When analyzing the GNSS position cyberspoofing, distinguishing between legitimate and altered signals is challenging, since both measurements generate fundamentally valid and realistic behaviours. Several core metrics, like C/No, horizontal speed, horizontal dilution of precision, and the total satellite count, remain completely unaltered during the attack.

The observable differences are small and primarily statistical. For example, the spoofed data exhibits an approximate 37.25% decrease in the 50% distribution radius. While real-world position states deviate consistently from their medians, the spoofed values tend to cluster closely around the median, interrupted only by regular, predictable spikes. This behaviour is reflected with higher spikes in the comparison between raw $\Delta pr/\Delta t$ and pseudorange rate. Despite these statistical variances, none is a standalone and sufficient indicator of a cyberspoofing attack, as we cannot definitively classify a signal based on these observations alone.

Because anomalies such as data spikes can occur in legitimate data, relying solely on GNSS data analysis for detection is insufficient: it is practically impossible to know a priori exactly how a receiver will behave in real-world signal propagation environments, like urban multipath scenarios. A first countermeasure could be to analyze the time series of the position, making the device store the last few seconds of positioning data to check for impossible jumps in location which exceed the velocity and acceleration capabilities of the receiver. However, while this is effective against abrupt position alterations, an attacker could still introduce a gradual drift to avoid detection and thus successfully conduct the attack. Consequently, the most robust countermeasure is cross-validation, relying on tertiary location services, such as cellular networks or Wi-Fi positioning, to establish a multi-system consensus. This approach exponentially increases the difficulty for an attacker, as successfully altering the target's location would require the simultaneous compromise of multiple independent positioning systems.

2.2.2 Delay Spoofing. The forms of delay spoofing are two:

i) Global delay: All the satellite signals are delayed. When a global delay spoofing attack is performed, the primary observation is an increase in pseudorange, since the attack introduces a larger difference between the signal's transmission and arrival time. In our case, the upper bound for pseudoranges went from 2.7×10^7 m to 3.024×10^9 m, following the application of a 10-second delay:

$$c \times 10s \approx 3 \times 10^8 m/s \times 10s = 3 \times 10^9 m$$

Such a delay is unrealistic for a genuine signal, and it was chosen only as a test. Beyond this, the remaining data does not show relevant deviations. Crucially, adding a uniform delay to all satellite measurements does not affect the final computed spatial position, but rather the clock bias estimation (Appendix Figure 8), which could be used to correct the internal time of the receiver. Therefore, while harmless to devices that rely on GNSS for pure localization, it is highly damaging to critical infrastructure relying on precise time synchronization of GNSS.

ii) Selective delay: Only a subset of signals belonging to a specific set of satellites is delayed. Here, the attack disrupts the geometric consistency of the signals, since only specific satellites appear further away. This makes it impossible for the Least Squares algorithm to find a physically correct intersection for the pseudoranges, resulting in jumps in the computed location. In a real-world scenario, the receiver would detect these discrepancies and discard the GNSS measurements entirely, enabling the attacker to conduct a successful denial of service of the PVT data.

To counter these attacks, the device must necessarily rely on alternative sources. For global delay attacks, cross-validation in the time domain is essential, relying on timekeeping services such as NTP accessed via cellular networks or Wi-Fi, to ensure a multi-system consensus. In the event of a selective delay DoS attack, the system must recognize the anomaly and fall back to tertiary systems for positioning and timing until GNSS integrity is restored.

2.3 Peculiar Situations

2.3.1 Indoor. Retracing the behaviour analyzed in the *Line of Sight* scenario, even in the *Indoor* one, the Samsung A51 and Redmi Note 7 outperformed the Google Pixel 7a, reporting an average variance for 50% of the distribution of 21.5m, 49.3m and 52.25m respectively with power saving enabled. Interestingly, the Samsung A51 showed no relevant differences between the non-power saving and power saving scenarios. The non-power save measurements for the Google Pixel 7a were incomplete, preventing any further research.

As was expected, this scenario provided the least accurate results overall (except the *Subway* one), which matches the impossibility of having line of sight with any satellites derived from being inside a building.

2.3.2 Subway. Four additional measurements are taken inside subway cars while travelling. As expected, generated logs are incomplete, likely because of an insufficient SNR for received signals.

Consequently, it is not possible to plot results or compute any position.

2.3.3 Car. Analysing GNSS measurements taken from inside a moving car in Turin and comparing them to a less obstructed scenario (e.g., *DVB-T Antenna* measurements), the most significant aspects to analyze are (i) the pseudoranges change, (ii) the number of locked satellites and (iii) the HDOP.

Since the receiver moves inside an urban canyon, the presence of obstacles would likely limit the reception of satellites close to the horizon. Moreover, said satellites should be characterized by a faster change in the pseudorange compared to satellites with higher elevations because of their increased radial velocity.

As depicted in Figure 2, the results were partially aligned with the expectations. Due to the presence of obstacles, the *Car* scenario in an urban canyon introduces the phenomenon of multipath, which both compromises the signal stability and causes frequent loss of lock. In Figure 5, the bottom plot shows spikes at the loss of lock points, which correspond to the geometric degradation of the constellation. The *DVB-T Antenna* scenario additionally enables getting better exposition. Indeed, in the upper plot of Figure 5, the number of satellites remains constant over time, resulting in an optimal geometric distribution.

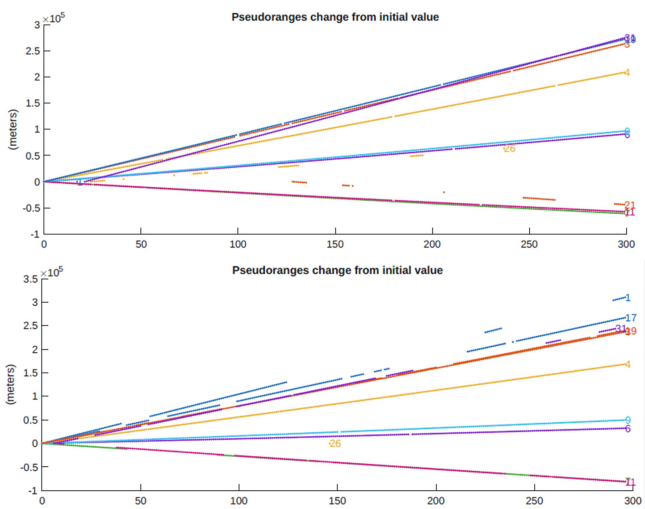

Figure 2: Comparison of pseudoranges change in *DVB-T Antenna* (top) and *Car* (bottom) scenario for the Samsung A51, power save disabled. The broken lines illustrate the real-world impact of urban canyons.

Through USCG [8], N2YO [4], and Stellarium [5] services, it was possible to determine the position of satellites during the measurements, and skyplots for the different samples are generated via a custom Python script available in the public repository. The skyplot relative to this scenario is depicted in Appendix Figure 7. PRN 26 identifies the satellite closest to the horizon among the ones that are received. However, its C/No is the lowest in the record (24dB/Hz), being received only briefly during the capture.

2.3.4 DVB-T Antenna. To analyze the effects of a potential out-of-band interference source, we collect GNSS measurements near the DVB-T antennas on *Colle della Maddalena (Turin)*. Unexpectedly, the positioning accuracy is not severely degraded, with 50% of the calculated positions falling within a 5.2-meter radius with the Google Pixel 7a, power saving enabled. However, this does not rule out the possibility of increased positioning errors if different GNSS receiver architectures are used. Nevertheless, the influence of the interference source is evident in Figure 3, which shows a uniform fluctuation of approximately 5 dB/Hz in the C/No values across all tracked signals.

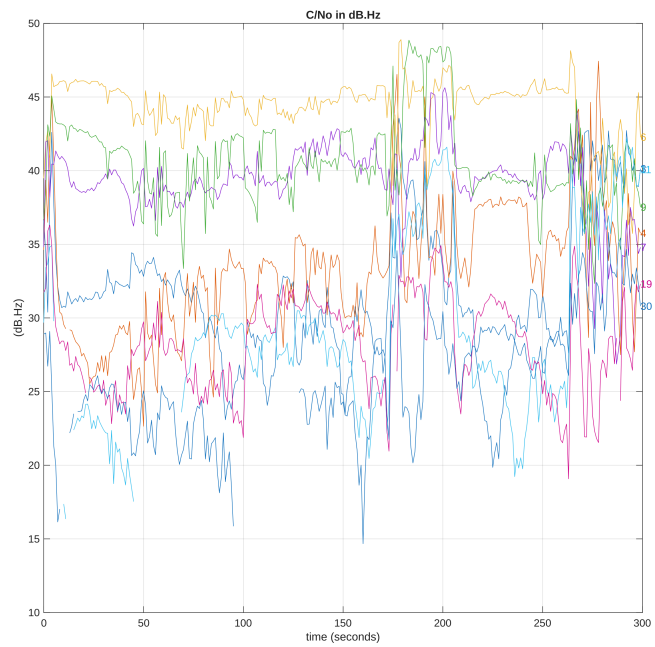

Figure 3: C/No uniform fluctuations close to DVB-T Antennas. Google Pixel 7a, power save enabled.

The skyplot of this measurement is depicted in Appendix Figure 6. Compared to the *Car* skyplot, the lowest satellite is PRN 01, now located 12.1 degrees lower than before. However, the lower amount of obstacles enables receiving it with a C/No around 27dB/Hz, 7dB/Hz higher than before. This, however, lasts for a very brief moment before the satellite sets over the horizon.

3 CONCLUSIONS

GNSSs are continuously exposed to interference and attacks. With this study, we demonstrated how the choice of GNSS receiver can have an effect on measurements and how obstacles limit low-elevation satellite reception. Furthermore, we explored how out-of-band interference can practically affect the signal quality and how cyberspoofing can subtly alter the positioning, discussing possible detection strategies.

Given that this work can be considered a preliminary study, future work must include a greater number of samples to estimate performance differences and better comprehend the real impact interference has. A more granular experimental setup would enable better isolation of the causes, such as moving subjects or weather conditions. Finally, further exploration of detection and mitigation techniques can be performed.

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A APPENDIX

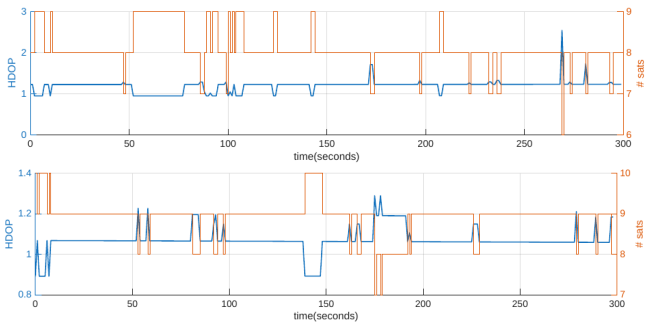

Figure 4: HDOP and satellites for the worst (top) and best (bottom) sample recorded by the Google Pixel 7a, power save disabled, in the *Line of Sight* scenario.

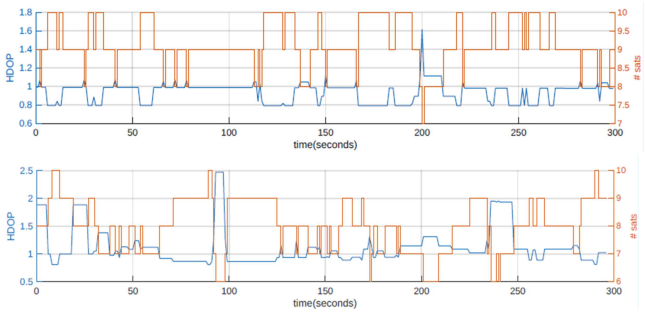

Figure 5: Comparison of HDOP and Number of Satellites in *DVB-T Antenna* (top) and *Car* (bottom) Scenario. Samsung A51, power save disabled.

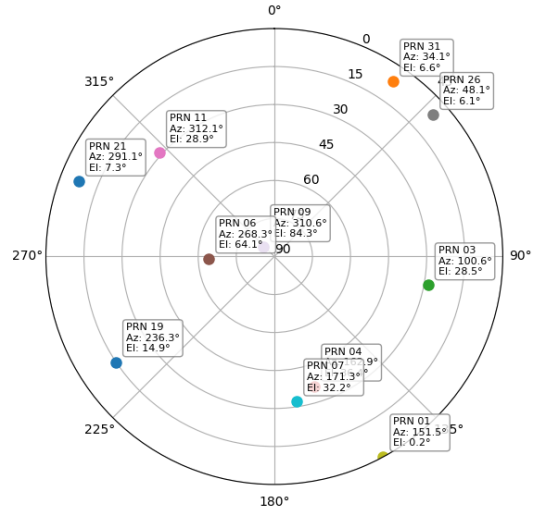

Figure 6: *DVB-T Antenna* measurement skyplot. Google Pixel 7a, power save enabled.

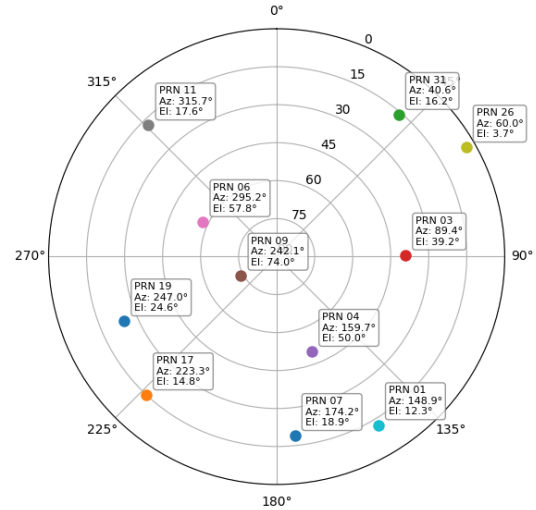

Figure 7: *Car* measurement skyplot. Samsung A51, power save disabled.

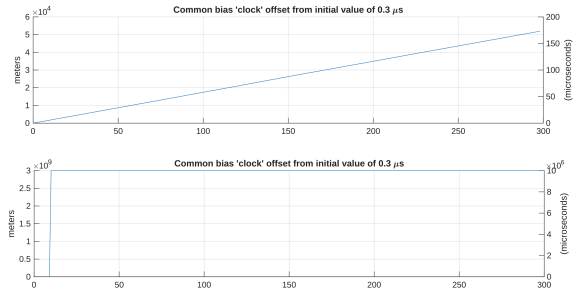

Figure 8: Clock bias variation when delay is added. Genuine (top) and Spoofed (bottom). Google Pixel 7a, power save disabled.